Supplementary Material

Contents

1. MEDLINE Search Strategy.....2
2. Summary of Study Characteristics.....3
3. Outcome of the Risk of Bias Assessment.....5
4. List of references for included studies 7

***Figure S1**. Search Strategy for MEDLINE Database, searched from inception till January 2021.*

Key concepts	**Concept 1-** Patient, Population **Patients with cardiac diseases (OR)**		**Concept 2 -** Intervention (or Exposure) **Exercise in Cardiac Rehabilitation (OR)**		**Concept 3 -** Comparison (or Control) if appropriate **Exercise intensity/Programs (OR)**		**Concept 4 -** Outcome **Aerobic capacity (OR)**
Free text terms / natural language terms	Myocard* near/5 isch*mi* Isch*mi* near/5 heart Myocard* near/5 infarct* Heart near/5 infarct* Angina Coronary near/5 disease* or bypass or thrombo* or angioplast* (Percutaneous next coronary near/2 interven* or revascular* Angioplast* coronary or arterial near/4 dilat* Endoluminal next repair* Stent* Pci or ptca Atherectom* Acute next coronary next syndrom*	**AND**	Rehabilitat* Physical* near/5 fit* or train* or therap* or activit* train* near/5 strength* or aerobic* or exercise* exercise* or fitness near/3 treatment or intervent* or program* Kinesiotherap* patient* near/5 educat* lifestyle or life-style near/5 interven* or program* or treatment* self near/5 manag* or care or motivate* Psychotherap* Psycholog* near/5 intervent* counselling or counseling behavior* or behaviour* near/5 modify or modificat* or therap* or change psycho-educat* or psychoeducat* motivat* near/5 (intervention or interv* health near/5 educat* psychosocial or psycho-social cognitive near/2 behav*	**AND**	Exercise Test/methods* High-Intensity Interval Training* HIIT HIIT or HIT Exercise intensit* Exercise next intensit* Exercise near/5 intensit* Exercise Intensity Program* intensit* near/5 interven* or program* or treatment* intensity* near/5 interven* or program* or treatment* intensit* near/5 method* intensit* near/5 exercise* intensit* near/5 program* Moderate continuous training Moderate-intensity continuous training moderate near/4 training MCT moderate continuous training near/5 methods* moderate continuous training near/5 standards* *rehabilitation near/5 exercise* *rehabilitation near/5 program* Home-based Home-based near/5 methods* Home-based near/5 satandards* Aerobic Exercises Anaerobic Exercises No exercise	**AND**	Heart Failure next mortality Heart Failure near/5 mortality Cardiovascular Capacity* Oxygen Consumption/physiology*
Controlled vocabulary terms / Subject terms	Intervention (MeSH) Coronary Artery Bypass (MeSH) Percutaneous Coronary Intervention (MeSH) Angioplasty (MeSH) Stents (MeSH) Atherectomy (MeSH)		Exercise Therapy (MeSH) Sports (MeSH) Physical Exertion (MeSH) Exercise (MeSH) Rehabilitation (MeSH) Physical Education and Training (MeSH) Patient Education as Topic (MeSH) Self Care (MeSH) Psychotherapy (MeSH) Counseling (MESH) Health Education (MeSH)		Exercise Tolerance (MeSH) High Intensity Interval Training/methods (MeSH) High Intensity Interval Training/standards (MeSH) Telerehabilitation (MeSH) Telerehabilitation/methods (MeSH) Telerehabilitation/standards (MeSH) Cardiac Rehabilitation/methods (MeSH) Cardiac Rehabilitation/standards (MeSH) High Intensity Interval Training/standards (MeSH)		Surveys and Questionnaires (MeSH) Cardiovascular Diseases/epidemiology (MeSH) Cardiovascular Diseases/mortality (MeSH) Morbidity/trends (MeSH) Motor Activity/physiology*(MeSH) Prevalence (MeSH) Risk Factors (MeSH) Sedentary Lifestyle (MeSH) Oxygen Consumption/physiology (MeSH) Predictive Value of Tests (MeSH) Reproducibility of Results (MeSH) Recovery of Function (MeSH)

***Table S2**. Summary of Study Characteristics.*

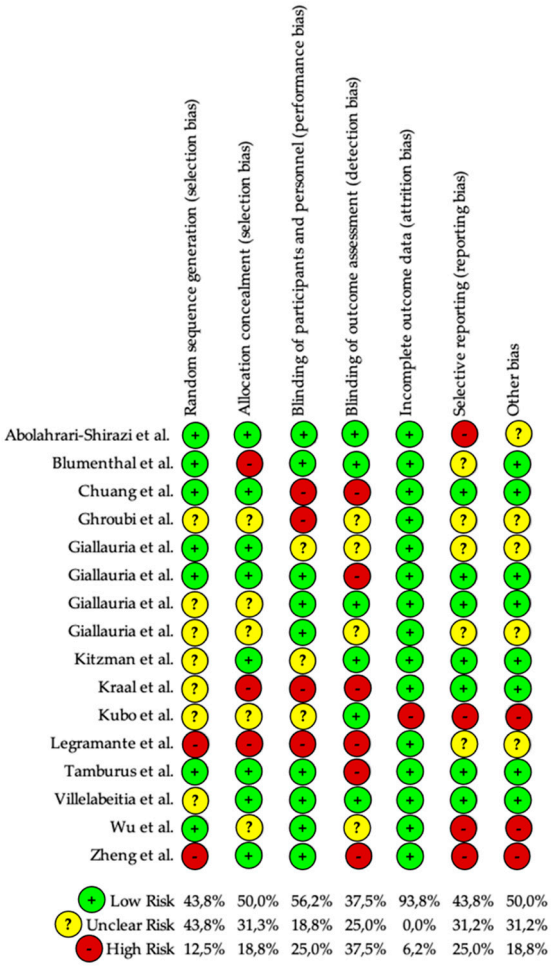
		Participant characteristics						Rehabilitation protocols								Primary Outcome	
Study	**Year**	**n (men)**	**Age**	**Conditions**	**Length**	**Freq.**	**Exercise duration**	**Intensity**	**Mode**	**Type**	**Sup.**	**Prog.**	**Res.**	**Comp.**	**ΔVO2R**	**ΔVO2AT**
Abolahrari-Shirazi et al.	2018	25 (NS)	56.76 (8.71)	PCI	7	3	W:5; A:30; C:5	40-70%VO2p	CY+AE+TM+RES	Cont	CL/H	Y	Y	Y	8.2	
		25 (NS)	57.64 (7.85)	PCI	7	3	W:5; A:45; C:5	40-70%VO2p	CY+AE+TM	Cont	CL/H	Y	N	Y	9.4	
		25 (NS)	57.32 (9.41)	PCI								N	N	Y	1.8	
Blumenthal et al.	2005	48 (31)	62±11	IHD EMI	16	3	W:10; A:35; C:10	70-85%HRR	WK	Cont	CL	N	N	N	1.9	
		44 (29)	63±12	IHD EMI	16	1			STM		CL	N	N	N	0.3	
		42 (32)	63±9	IHD EMI								N	N	N	-0.3	
Chuang et al.	2005	17 (15)	64±8	CABG	12	2	A:30	70-80%HRp, 60-70%VO2p, RPE 11-15	TM	Cont	CL	N	N	N	4.76	
		15 (13)	69±12	CABG								N	N	N	1.72	
Ghroubi et al.	2013	16 (NS)	59±6	CABG	8	3	W:10; A:20; C:10	70%HRR	CE	Cont	CL	N	N	N	1.70	
		16 (NS)	59±2	CABG	8	3	W:5; A:20;	70%HRR (20-30% Peak torque)	RES	Cont/Int	CL	N	Y	N	4.00	
Giallauria et al.	2006	22 (16)	55±8	MI	12	3	W:5; A:30; C:5	60-85%VO2p	CE+CY	Cont	CL/H	Y	N	N	4.2	
		22 (17)	54±10	MI	12	3	W:5; A:30; C:5	60%VO2p	CE	Cont	CL	Y	N	N	3.8	
Giallauria et al.	2009	26 (2)	58±8	MI	12	3	W:5; A:30; C:5	60-70%VO2p	CE	Cont	CL	N	N	N	4.3	2.3
		26 (2)	57±10	MI								N	N	N	-2	-1.7
Giallauria et al.	2011	37 (28)	61±7	MI	26	3	W:5; A:30; C:5	60-70%VO2p	CE	Cont	CL	N	N	Y	4	
		26 (23)	52±10	MI	26							N	N	N	1	
Giallauria et al.	2013	25 (22)	54±7	MI	26	3	W:5; A:30; C:5	60-70%VO2p	CE	Cont	CL	N	N	N	4.00	
		21 (18)	54±9	MI								N	N	N	1.00	
Kitzman et al.	2013	32 (NS)	70±7	FMD CAS	16	3	W:10; A:20(WK)+20(E); C:10	40/50-70%HRR	WK, AE, CE	Cont	CL	Y	N	N	1.6	

		31 (NS)	70±7	FMD CAS	16	2						N	N	N	-0.2
Kraal et al.	2013	25 (21)	56±9	PCI CABG	12	2	A:45-60	70-85%HRp	TM, CE	Cont	CL	N	N	N	2.40
		25 (22)	61±8	PCI CABG	12	2	A:45-60	70-85%HRp	TM, CE	Cont	CL/H	N	N	N	3.20
Kubo et al.	2004	24 (21)	59±12	MI	12	3	A:320	60-70%HRp	TM, CE	Cont	CL	N	N	N	2.9
		24 (17)	62±12	MI	12							N	N	N	-0.3
Legramante et al.	2017	43 (NS)	60±9	CABG	2	6x2	A:30	75-85% HRp	WK, CAL, CE	Cont	CL	Y	Y	N	2.6
		39 (NS)	58±8	CABG	2	6x2	A:30	75-85% HRp	WK, CAL	Cont		Y	N	N	0.9
Tamburus et al.	2016	15 (NS)	57±7	CAD	16	3	W:10; A: 30-40; C:10	70-110% VO2VAT	CE	Int	CL	Y	N	N	1.51
		17 (NS)			16							N	N	N	-1.86
		15 (NS)	57±7	None	16	3	W:10; A: 30-40; C:10	70-110% VO2VAT	CE	Int	CL	Y	N	N	1.95
		17 (NS)		None	16							N	N	N	-1.43
Villelabeitia et al.	2017	37 (29)	58±11	CAD	8	3	W:5-12; A:15-30; C:5-13	104.5% ± 22.2% VO2p (1ºmonth) and 134.5% ± 29.7% VO2p (2ºmonth)	CE	Int	CL	Y	N	N	4.5
		36 (33)	58±11	CAD	8	3	W:5-12; A:15-30; C:5-13	64.2% ± 8.5% VO2p (1ºmonth) and 69.5% ± 8.7% VO2p (2ºmonth)	CE	Cont	CL	Y	N	N	2.46
Wu et al.	2006	18 (NS)	63±7	CABG	12	3	W:10; A:30-60; C:10	60-85 %HRp	TM, CE	Cont	CL	N	N	N	8.50
		18 (NS)	61±8	CABG	12	3	W:10; A:30-60; C:10	60-85 %HRp, RPE 11-13	WK	Cont	H	N	N	N	6.50
		18 (NS)	62±10	CABG								N	N	N	3.50
Zheng et al.	2008	27 (NS)	NS	MI	26	3	W:15; A:30; C:15	100% AT	CE	Cont	CL	N	N	N	3.10
		30 (NS)	NS	MI	26							N	N	N	0.3

NS, not stated/missing. n(men) presented as the sample size (number of men). Age presented as mean±SD years. Conditions: MI, myocardial infarction. CABG, coronary artery bypass graft. PCI, percutaneous coronary intervention. CAD, coronary artery disease. IHD, ischemic heart disease. EMI, exercise-induced myocardial ischemia. FMD, endothelial-dependent flow-mediated arterial dilation. CAS: carotid artery stiffness. Rehabilitation protocols: Length presented as no. of weeks. Frequency (Freq.) presented as sessions per week. Exercise Duration: presented as minutes per session: W, warm-up. A: aerobic component (interval programs presented as interval x duration). C, cool-down. SMT: stress management training. R, recovery. wk, week. Intensity: %HRp, % peak heart rate. %HRR, % heart rate reserve. %VO_2peak, % peak oxygen uptake. %AT/VT, % of anaerobic/ventilatory threshold. RPE, rating of perceived exertion. Mode: TM, treadmill. CE, cycle ergometer. AE, arm ergometer. RES, resistance training protocol. WK, walking/jogging. CY, cycling, training. CAL: Calisthenics. Type: Cont, continuous training. Int, interval. Supervision (Sup.), level of

monitoring/supervision: CL, clinic-based. H, home-based. Progressive (Prog.), whether aerobic exercise intensity was re-evaluated during the program: Y, yes. N, no. Resistance exercises (Res.): Y, yes. N, no. Comprehensive rehabilitation (Comp.), exercise training plus education and risk factor management: Y, yes; N, No. Outcomes: $\Delta V\dot{O}_2R$, change in relative VO_2peak (presented as $mL \cdot kg^{-1} \cdot min^{-1}$).

***Figure S3**. Outcomes of the risk of bias assessment.*



For references, see Supplementary Material S4.

S4. *List of references for included studies*

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